

Project Repositories Summary

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Project Repositories Summary

Workspace Inventory Report

/home/markt/proj

Generated from local repository inspection

This directory contains 8 Git repositories. The notes below are based on each repo's local `README.md`, top-level files, and current Git status as inspected on 2026-04-22.

At a glance

- **IP-ucie-rdi-to-pcie-pipe**: UCIE RDI to PCIe PIPE bridge RTL. Stack: SystemVerilog, Verilator, vendor simulators. State: `main`, dirty.
- **ucie-cxl-bridge**: UCIE to CXL bridge RTL. Stack: Verilog/SystemVerilog, Icarus, Verilator, formal. State: `main`, clean.
- **IP-axi-to-2apbs**: AXI to dual-APB4 bridge RTL. Stack: Verilog, Icarus. State: `main`, clean.
- **chi-to-bow-bridge**: CHI to BoW bridge starter. Stack: Verilog, Cocotb, Icarus, Pandoc. State: `main`, dirty.
- **axi4_to_dfi_ddr**: AXI4 to DFI / DDR bridge RTL. Stack: Verilog, Icarus, Verilator, Yosys, Pandoc. State: `main`, clean.
- **riscv_test_asm_qemu**: RISC-V cross-compile and QEMU experiments. Stack: RISC-V GNU toolchain, QEMU. State: `master`, clean.
- **MiT_capstone_beetle_kill**: Forest bark beetle object detection. Stack: Python, PyTorch, pytest. State: `codex/review-improvements`, dirty.
- **pc-wsl-github-starter**: WSL + GitHub starter workflow. Stack: Python, unittest, GitHub Actions. State: `main`, ahead 1, dirty.

Repository notes

IP-ucie-rdi-to-pcie-pipe

- Purpose: production-style SystemVerilog bridge from UCIE 1.0 RDI to PCIe Gen4 PIPE.
- Highlights: dual-clock CDC handling, per-lane elastic FIFOs, CRC32 support, assertions, and Verilator-oriented simulation flow.
- Current state: active local edits in RTL, testbench, docs, and build files; untracked `.github/`, `obj_dir/`, and `sim_top.sv`.
- Last commit: `0ff1080` on 2026-04-01, "Fix diag format".

ucie-cxl-bridge

- Purpose: experimental UCIE Adapter Layer to CXL.io / CXL.cache / CXL.mem bridge.
- Highlights: baseline dual-FIFO streaming shell, simulation CI, optional Verilator lint, and bounded formal checks on `sync_fifo`.
- Current state: clean working tree on `main`.
- Last commit: `a33af7b` on 2026-04-13, improving CI formal path handling.

IP-axi-to-2apbs

- Purpose: AXI to 2x APB4 bridge RTL with self-checking testbenches.
- Highlights: separate simple and burst bridge variants, plus APB read wait-state test sweeps through `make`.
- Current state: clean working tree on `main`.

- Last commit: 0cc3a3c on 2026-04-20, fixing APB timing and simulation flow.

chi-to-bow-bridge

- Purpose: starter CHI-to-BoW bridge that packetizes simplified CHI requests into BoW flits and reconstructs responses.
- Highlights: Cocotb-based verification, documentation build flow, waveform generation, and protocol guardrails for invalid response patterns.
- Current state: local edits in README, design spec, RTL, and Cocotb tests.
- Last commit: 8d6e507 on 2026-04-21, merge of the next milestone branch.

axi4_to_dfi_ddr

- Purpose: AXI4 slave to JEDEC DFI-style bridge for a DDR PHY / memory-controller path.
- Highlights: async FIFOs for CDC, open-page scheduling, parameterized timing, smoke tests, elaboration-fail checks, optional Verilator lint, Yosys synthesis/formal hooks, and generated design docs.
- Current state: clean working tree on **main**.
- Last commit: ae3401f on 2026-04-12, adding elaboration guards for MC timing and DFI init cycles.

riscv_test_asm_qemu

- Purpose: small RISC-V bring-up / learning repo for assembling, compiling, running, and disassembling simple programs under QEMU.
- Highlights: Make targets for assembly and C++, targeted objdump outputs (**_start**, **main**, source-interleaved **main**), and simple output-based tests.
- Contents: **hello.s**, **howdy.cpp**, generated binaries/objects, and a Makefile; there is no repo README yet.
- Current state: clean working tree on **master**.
- Last commit: 438e894 on 2026-04-06, adding source-interleaved main disassembly.

MiT_capstone_beetle_kill

- Purpose: PyTorch object-detection pipeline for aerial / oblique forest bark beetle survey imagery with Pascal-VOC-style XML annotations.
- Highlights: Faster R-CNN training, inference, evaluation metrics, smoke tests, pytest-based tests, threshold tuning scripts, and dataset/process docs.
- Current state: on branch **codex/review-improvements** with untracked local temp/test artifacts (**.tmp_***) plus **.codex**.
- Last commit: 8c2ea0b on 2026-04-20, adding automated tests and stricter smoke assertions.

pc-wsl-github-starter

- Purpose: minimal starter repo for a Windows PC + WSL + GitHub workflow.

- Highlights: tiny Python app, unittest example, WSL-friendly Git config files, and a GitHub Actions workflow.
- Current state: `main` is ahead of `origin/main` by 1 commit, with local edits to `Makefile` and untracked `.codex` and `docs/`.
- Last commit: `d8bb9c9` on 2026-04-03, adding a local workflow `Makefile`.

Overall observations

- Most repositories in this directory are hardware / verification projects focused on RTL bridges and CDC-heavy interfaces.
- `MiT_capstone_beetle_kill` is the main non-RTL project; it is a Python/PyTorch ML pipeline.
- Four repos currently have local uncommitted changes: `IP-ucie-rdi-to-pcie-pipe`, `chi-to-bow-bridge`, `MiT_capstone_beetle_kill`, and `pc-wsl-github-starter`.
- `riscv_test_asm_qemu` is the only repo here without its own README, so this top-level summary is the main written orientation for it right now.